Help desk homelab starter checklist

Use disposable systems, invented data and a verified recovery path. Work from the equipment you already own. These are practice plans, not evidence of completed tests.

01. Start with the computer you own

Check hardware requirements and available disk/RAM for one small disposable VM. Back up important host files before changing storage or firmware settings. Follow the current virtualization and evaluation-license documentation.

02. Separate practice from real data

Use disposable VMs, spare drives or a spare machine for drills. Label disks clearly. Never practice deletion, partitioning or restoration on a daily-use computer or the only copy of anything important.

03. Build an isolated practice network

Prefer a host-only or deliberately isolated virtual network for DHCP, DNS and domain experiments. Do not bridge a practice DHCP server onto a home or workplace LAN. Record which adapter reaches which network.

04. Add storage only for a specific requirement

Choose an external drive or adapter only after checking interface, capacity and power requirements. One external drive is not a complete backup strategy. Verify another independent copy of important files.

05. Verify remote access before hardening SSH

Use a local console as a recovery path. Create and verify a separate key-authenticated session before disabling password login. Check that the intended administrator can still connect; keep the console open until tested.

06. Test power protection without risking working data

Choose UPS capacity from the actual load and manufacturer guidance. Practice shutdown integration on a disposable system first. Keep printers, heaters and other unsuitable loads off battery-backed outlets.

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07. Restore a disposable VM, not your laptop

Create a disposable VM with test files. Back it up to a different target, verify that backup and bootable recovery media, and restore into a second disposable VM or spare virtual disk. Compare the restored files before trying any destructive exercise.

08. Practice DNS failures inside the VM

Snapshot the VM and record its working network settings. Change only that VM's DNS configuration, then inspect address, gateway and DNS results. Restore the known-good settings and verify resolution afterward.

09. Practice permissions with invented users and files

Create test accounts and dummy folders in the lab. Record expected access before changing permissions, test allowed and denied cases, and restore the baseline. Never use real customer or employer data.

10. Keep a ticket for each drill

Write the symptom, reproduction, expected result, diagnosis, change, validation and rollback. Include actual output only when you captured it. Redact credentials, private addresses and personal information before publishing.

11. Buy the next device when a measured limit blocks you

Do not buy a rack, NAS, managed switch or faster network merely to start. Measure RAM, storage and network needs first. VLAN practice may justify a managed switch later; the skill comes before the shopping list.

12. Finish with a working baseline

Undo the intentional break, check networking and services, confirm backups and recovery access, and record remaining faults honestly. A screenshot of a dashboard is not proof that restoration or access controls work.

Source checks and next steps

Microsoft evaluation software: check current license and requirements
https://www.microsoft.com/en-us/evalcenter/

Recovery: protect the source and save recovered data elsewhere
https://support.microsoft.com/en-us/windows/experience/backup-recovery/windows-file-recovery